```
!rm -rf SLD_MAHE
!git clone https://<private_token_hidden>@github.com/madhavasthana-oss/SLD_MAHE
```

```
Cloning into 'SLD_MAHE'...
remote: Enumerating objects: 190, done.
remote: Counting objects: 100% (190/190), done.
remote: Compressing objects: 100% (134/134), done.
remote: Total 190 (delta 103), reused 134 (delta 51), pack-reused 0 (from 0)
Receiving objects: 100% (190/190), 301.41 KiB | 12.56 MiB/s, done.
Resolving deltas: 100% (103/103), done.
```

```
%cd /content/SLD_MAHE/backend
```

```
/content/SLD_MAHE/backend
```

```
import kagglehub

# Download latest version
path = kagglehub.dataset_download("grassknoted/asl-alphabet")

print("Path to dataset files:", path)
```

```
Using Colab cache for faster access to the 'asl-alphabet' dataset.
Path to dataset files: /kaggle/input/asl-alphabet
```

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
!python training_pipeline.py \
--root_dir /kaggle/input/asl-alphabet/asl_alphabet_train/asl_alphabet_train \
--save_to /content/drive/MyDrive/SLD \
--epochs 3 \
--batch_size 128
```

```
Downloading: "https://download.pytorch.org/models/resnet50-11ad3fa6.pth" to /root/.cache/torch/hub/checkpoints/resnet
100% 97.8M/97.8M [00:00<00:00, 229MB/s]
<<<< Dataset splits >>>>
  train : 60899
  val   : 17400
  test  : 8701
wandb: (1) Create a W&B account
wandb: (2) Use an existing W&B account
wandb: (3) Don't visualize my results
wandb: Enter your choice: 2
wandb: You chose 'Use an existing W&B account'
wandb: Logging into https://api.wandb.ai. (Learn how to deploy a W&B server locally: https://wandb.me/wandb-server)
wandb: Create a new API key at: https://wandb.ai/authorize?ref=models
wandb: Store your API key securely and do not share it.
wandb: Paste your API key and hit enter:
wandb: ERROR Invalid API key: API key must have 40+ characters, has 1.
wandb: (1) Create a W&B account
wandb: (2) Use an existing W&B account
wandb: (3) Don't visualize my results
wandb: Enter your choice: 2
wandb: You chose 'Use an existing W&B account'
wandb: Logging into https://api.wandb.ai. (Learn how to deploy a W&B server locally: https://wandb.me/wandb-server)
wandb: Create a new API key at: https://wandb.ai/authorize?ref=models
wandb: Store your API key securely and do not share it.
wandb: Paste your API key and hit enter:
wandb: No netrc file found, creating one.
wandb: Appending key for api.wandb.ai to your netrc file: /root/.netrc
wandb: Currently logged in as: madhavasthana (madhavasthana-manipal) to https://api.wandb.ai. Use `wandb login --relo
ccwandb: ⣻ setting up run 9mb3xxz8 (0.1s)
wandb: ⣽ setting up run 9mb3xxz8 (0.1s)
wandb: Tracking run with wandb version 0.25.1
wandb: Run data is saved locally in /content/SLD_MAHE/backend/wandb/run-20260414_062159-9mb3xxz8
wandb: Run `wandb offline` to turn off syncing.
wandb: Syncing run deep-dust-23
wandb: ⭐️ View project at https://wandb.ai/madhavasthana-manipal/Sign%20language%20detection
wandb: 🚀 View run at https://wandb.ai/madhavasthana-manipal/Sign%20language%20detection/runs/9mb3xxz8
```

```
<<<< Starting training for 3 epochs >>>>

  epoch 1/3 | train_loss: 3.0098 | train_acc: 26.09% | val_loss: 2.2431 | val_acc: 64.41% | lr: 0.002000
<<<< New best model saved — val_acc: 64.41% >>>>
  epoch 2/3 | train_loss: 0.7375 | train_acc: 86.05% | val_loss: 0.0944 | val_acc: 98.29% | lr: 0.003000
<<<< New best model saved — val_acc: 98.29% >>>>
  epoch 3/3 | train_loss: 0.0474 | train_acc: 99.03% | val_loss: 0.0099 | val_acc: 99.86% | lr: 0.004000
<<<< New best model saved — val_acc: 99.86% >>>>

<<<< Training complete >>>>
<<<< Loading best checkpoint for final evaluation >>>>

<<<< Final Test Results >>>>
  test_loss : 0.0096
  test_acc  : 99.80%
wandb: ⣾ updating run metadata (0.0s)
wandb: ⣽ updating run metadata (0.0s)
wandb: ⣻ updating run metadata (0.0s)
wandb: ⢿ updating run metadata (0.0s)
wandb: ⡿ uploading config.yaml 2.9KB/2.9KB (0.2s)
wandb: ⣟ uploading config.yaml 2.9KB/2.9KB (0.2s)
```

Start coding or generate with AI.